Prior Sensitivity Analysis

2026-04-24

Setup

```r
library(UsingR)
```

```
## Warning: package 'UsingR' was built under R version 4.4.3

## Loading required package: MASS

## Loading required package: HistData

## Warning: package 'HistData' was built under R version 4.4.3

## Loading required package: Hmisc

## Warning: package 'Hmisc' was built under R version 4.4.3

##
## Attaching package: 'Hmisc'

## The following objects are masked from 'package:base':
##
##     format.pval, units
```

```r
library(Bolstad)
```

```
##
## Attaching package: 'Bolstad'

## The following objects are masked from 'package:stats':
##
##     IQR, sd, var
```

```r
babies_df <- babies
```

```r
# Remove all observations with missing or unknown values
babies_df = subset(babies_df, wt != 999 & parity != 99 & age != 99 &
                     ht != 99 & wt1 != 999 & smoke != 9 & number != 99 & number != 98)
```

Model

We fit a multiple linear regression of birth weight (oz) on six maternal predictors:

- gestation (days): longer pregnancies produce heavier babies
- ht (inches): taller mothers tend to have larger babies
- wt1 (lbs): heavier pre-pregnancy mothers tend to have heavier babies
- number: ordinal measure of cigarettes smoked per day (0=never, ..., 8=60+)
- parity: 0 = first birth, 1 = previous birth(s)
- age (years): maternal age at delivery

```
wt ~ gestation + ht + wt1 + number + parity + age
```

There are 7 coefficients total (including intercept).

Prior Specifications

We compare four normal prior specifications. The prior mean b0 encodes our expectation about the direction and rough magnitude of each effect before seeing the data. The prior variance V0 encodes our uncertainty about those expectations. We use three prior mean vectors:

Prior 1 – Diffuse, zero mean (baseline) No prior knowledge assumed. Mean of zero on all coefficients, very large variance. This is the same prior structure used in the smoking-only model.

Prior 2 – Diffuse, zero mean, moderate variance Same zero-mean assumption but variance scaled down to 10 * I. This provides mild regularization without encoding any directional beliefs.

Prior 3 – Informative, research-backed means, diffuse variance Prior means are derived from published literature on each predictor's relationship with birth weight. Variance remains diffuse so the data can still move the posterior substantially. This prior asks: if we encode what existing research suggests about the direction and rough magnitude of each effect, do our posteriors change?

Prior mean derivations from literature review: - Intercept: set to 0 (absorbed into diffuse variance)

- gestation: Previous studies report median birth weight increments of ~140g per gestational week from a US national reference sample data, independent of the CHDS. Converting to oz/day: 140g / 28.35g/oz / 7 days ~= 0.70 oz/day. Source: Oken et al. (2003). BMC Pediatrics, 3:6. doi:10.1186/1471-2431-3-6

- ht: consistent positive association across studies for height of mother and birthweight of child ; a 5cm increase in maternal height is associated with increases in newborn size about 47g. Thus, we encode a positive prior of .84 oz/inch based on prior studies data. Source: Thame et al. (2004)

- wt1: positive association of weight of mother and birthweight. A prior study reports about increase in birthweight of 37g/5kg of mothers weight. Thus, we encode a positive prior of 0.12 oz/lb. Source: Thame et al. (2004)

- number: Some researchers find light smokers (1-9 cigs/day) deliver infants 162g lighter and heavy smokers 226g lighter than non-smoking mothers. The number variable is ordinal (0=never, ..., 9=60+ cigs/day); the light smoker range spans roughly steps 2-3, giving ~65g per step, and the heavy smoker range gives ~40g per step. Converting the light-smoker estimate: 65g / 28.35 ~= -2.29 oz/step; heavy-smoker estimate: 40g / 28.35 ~= -1.41 oz/step. We use -1.8 oz/step as the prior mean, reflecting the midpoint weighted toward the more conservatively estimated heavy-smoker slope. Source: Juarez S, Merlo J (2013). doi:10.1101/2020.10.30.20222844

- parity: The babies dataset codes parity as binary: 0 = first birth, 1 = prior birth(s). So the relevant comparison is first-born vs any subsequent birth. From a previous study we know that Second-born are ~130g heavier than first-born and Third-or-later arre ~180g heavier than first-born. Thus if we assume a rough 60/40 split between second and third-or-later births among non-first births tgen$0.6 \times 130 + 0.4 \times 180 = 78 + 72 = 150$g increase for giving prior birth thus 150g / 28.35 = ~5.3 oz Source: Bohn et al. (2021)
- age: Prior studies shows that modest positive effect of age up to early 30s on birthweight and then reverse effects of ages. Thus we encode a zero prior. Source: Swamy et al. (2012)

Prior 4 – Informative means, tighter variance on weaker predictors Same research-backed means as Prior 3, but variance is tightened on all of them (variance = 10 rather than 1e4).

```
# --- Prior means ---

# Prior 1 & 2: zero mean (no prior belief)
b0_zero = rep(0, 7)

# Prior 3 & 4: research-backed means
# Order: intercept, gestation, ht, wt1, number, parity, age
b0_informed = c(0,      # intercept: no prior
                0.7,    # gestation: ~0.7 oz/day (143g/week converted)
                .84,    # ht: positive, ~1 oz/inch
                .12,  # wt1: small positive, ~0.05 oz/lb
                -1.8,   # number: negative, ~-1.5 oz per category step
                5.3,    # parity: positive, ~4 oz for non-first birth
                0)    # age: small positive, ~0.1 oz/year

# --- Prior variances ---

# Prior 1: Diffuse (baseline)
V0_diffuse = 1e4 * diag(rep(1, 7))

# Prior 2: Moderate zero-mean
V0_moderate = diag(c(1e4, 10, 10, 10, 10, 10, 10))

# Prior 3: Diffuse variance, informed means
V0_diffuse_informed = 1e4 * diag(rep(1, 7))

# Prior 4: Informed means, tighter variance on all of them
# Order: intercept, gestation, ht, wt1, number, parity, age
V0_tight_weak = diag(c(1e4,    # intercept: diffuse
                       10,    # gestation: diffuse
                       10,    # ht: tight
                       10,    # wt1: tight
                       10,    # number: tight
                       10,      # parity: tight
                       10))     # age: diffuse as not much information
```

```
fit1 = bayes.lm(wt ~ gestation + ht + wt1 + number + parity + age,
                data   = babies_df,
                center = FALSE,
                prior  = list(b0 = b0_zero,     V0 = V0_diffuse))
```

```
## Warning in model.matrix.default(mt, mf, contrasts): non-list contrasts argument
## ignored
```

```
fit2 = bayes.lm(wt ~ gestation + ht + wt1 + number + parity + age,
                data   = babies_df,
                center = FALSE,
                prior  = list(b0 = b0_zero,      V0 = V0_moderate))
```

```
## Warning in model.matrix.default(mt, mf, contrasts): non-list contrasts argument
## ignored
```

```
fit3 = bayes.lm(wt ~ gestation + ht + wt1 + number + parity + age,
                data   = babies_df,
                center = FALSE,
                prior  = list(b0 = b0_informed, V0 = V0_diffuse_informed))
```

```
## Warning in model.matrix.default(mt, mf, contrasts): non-list contrasts argument
## ignored
```

```
fit4 = bayes.lm(wt ~ gestation + ht + wt1 + number + parity + age,
                data   = babies_df,
                center = FALSE,
                prior  = list(b0 = b0_informed, V0 = V0_tight_weak))
```

```
## Warning in model.matrix.default(mt, mf, contrasts): non-list contrasts argument
## ignored
```

```
# bayes.lm strips names from coefficients via as.vector(); restore them
coef_names = c("(Intercept)", "gestation", "ht", "wt1", "number", "parity", "age")
names(fit1$coefficients) = coef_names
names(fit2$coefficients) = coef_names
names(fit3$coefficients) = coef_names
names(fit4$coefficients) = coef_names
```

Posterior Summaries

```
cat("=== Prior 1: Diffuse zero-mean (baseline) ===\n")
```

```
## === Prior 1: Diffuse zero-mean (baseline) ===
```

```
summary(fit1)
```

```
##
## Call:
## bayes.lm(formula = wt ~ gestation + ht + wt1 + number + parity +
##     age, data = babies_df, center = FALSE, prior = list(b0 = b0_zero,
##     V0 = V0_diffuse))
##
```

```
## Residuals:
##     Min      1Q  Median      3Q     Max
## -69.068 -10.577   0.172  10.978  52.923
##
## Coefficients:
##              Posterior Mean  Std. Error  t value
## (Intercept)   2.212e+01       1.839e+02   1.203e-01
## gestation     1.149e-02       4.811e-05   2.388e+02
## ht            1.375e+00       5.102e-02   2.694e+01
## wt1           5.920e-02       7.737e-04   7.651e+01
## number       -1.540e+00       5.572e-02  -2.763e+01
## parity        6.029e-02       9.808e-02   6.147e-01
## age           3.728e-02       1.064e-02   3.504e+00
## ---
## Prior Coefficients:
## (Intercept)    gestation           ht           wt1        number        parity
##           0            0            0             0             0             0
##         age
##           0
##
## Prior Covariance Matrix:
##              (Intercept)  gestation  ht     wt1    number  parity  age
## (Intercept)  10000            0          0      0      0       0       0
## gestation        0        10000          0      0      0       0       0
## ht               0            0      10000      0      0       0       0
## wt1              0            0          0  10000      0       0       0
## number           0            0          0      0  10000       0       0
## parity           0            0          0      0      0   10000       0
## age              0            0          0      0      0       0   10000
```

```r
cat("\n=== Prior 2: Moderate zero-mean ===\n")
```

```
##
## === Prior 2: Moderate zero-mean ===
```

```r
summary(fit2)
```

```
##
## Call:
## bayes.lm(formula = wt ~ gestation + ht + wt1 + number + parity +
##      age, data = babies_df, center = FALSE, prior = list(b0 = b0_zero,
##      V0 = V0_moderate))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -69.035 -10.603   0.185  10.984  52.945
##
## Coefficients:
##              Posterior Mean  Std. Error  t value
## (Intercept)   2.255e+01       1.831e+02   1.232e-01
## gestation     1.150e-02       4.811e-05   2.390e+02
## ht            1.367e+00       5.075e-02   2.693e+01
## wt1           5.964e-02       7.727e-04   7.718e+01
```

```
## number       -1.530e+00       5.541e-02  -2.762e+01
## parity        5.884e-02       9.710e-02   6.060e-01
## age           3.737e-02       1.060e-02   3.525e+00
## ---
## Prior Coefficients:
## (Intercept)    gestation          ht          wt1       number       parity
##           0            0           0            0            0            0
##         age
##           0
##
## Prior Covariance Matrix:
##             (Intercept)  gestation  ht     wt1    number  parity  age
## (Intercept)  10000           0          0      0       0       0       0
## gestation        0          10          0      0       0       0       0
## ht               0           0         10      0       0       0       0
## wt1              0           0          0     10       0       0       0
## number           0           0          0      0      10       0       0
## parity           0           0          0      0       0      10       0
## age              0           0          0      0       0       0      10
```

```
cat("\n=== Prior 3: Informed means, diffuse variance ===\n")
```

```
##
## === Prior 3: Informed means, diffuse variance ===
```

```
summary(fit3)
```

```
##
## Call:
## bayes.lm(formula = wt ~ gestation + ht + wt1 + number + parity +
##      age, data = babies_df, center = FALSE, prior = list(b0 = b0_informed,
##      V0 = V0_diffuse_informed))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -69.068 -10.578   0.172  10.978  52.923
##
## Coefficients:
##             Posterior Mean  Std. Error  t value
## (Intercept)   2.212e+01       1.839e+02   1.203e-01
## gestation     1.149e-02       4.811e-05   2.388e+02
## ht            1.375e+00       5.102e-02   2.694e+01
## wt1           5.920e-02       7.737e-04   7.651e+01
## number       -1.540e+00       5.572e-02  -2.763e+01
## parity        6.034e-02       9.808e-02   6.152e-01
## age           3.727e-02       1.064e-02   3.503e+00
## ---
## Prior Coefficients:
## (Intercept)    gestation          ht          wt1       number       parity
##        0.00         0.70        0.84         0.12        -1.80         5.30
##         age
##        0.00
##
```

```
## Prior Covariance Matrix:
##             (Intercept) gestation    ht   wt1 number parity   age
## (Intercept)       10000         0     0     0      0      0     0
## gestation             0     10000     0     0      0      0     0
## ht                    0         0 10000     0      0      0     0
## wt1                   0         0     0 10000      0      0     0
## number                0         0     0     0  10000      0     0
## parity                0         0     0     0      0  10000     0
## age                   0         0     0     0      0      0 10000
```

```r
cat("\n=== Prior 4: Informed means, tight variance on age/parity ===\n")
```

```
##
## === Prior 4: Informed means, tight variance on age/parity ===
```

```r
summary(fit4)
```

```
##
## Call:
## bayes.lm(formula = wt ~ gestation + ht + wt1 + number + parity +
##     age, data = babies_df, center = FALSE, prior = list(b0 = b0_informed,
##     V0 = V0_tight_weak))
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -69.46 -10.64   0.16  10.98  52.90
##
## Coefficients:
##             Posterior Mean  Std. Error   t value
## (Intercept)   2.227e+01      1.831e+02   1.216e-01
## gestation     1.146e-02      4.811e-05   2.382e+02
## ht            1.376e+00      5.075e-02   2.710e+01
## wt1           5.873e-02      7.727e-04   7.601e+01
## number       -1.541e+00      5.541e-02  -2.781e+01
## parity        1.109e-01      9.710e-02   1.142e+00
## age           2.875e-02      1.060e-02   2.712e+00
## ---
## Prior Coefficients:
## (Intercept)   gestation          ht         wt1      number      parity
##        0.00        0.70        0.84        0.12       -1.80        5.30
##         age
##        0.00
##
## Prior Covariance Matrix:
##             (Intercept) gestation    ht   wt1 number parity   age
## (Intercept)       10000         0     0     0      0      0     0
## gestation             0        10     0     0      0      0     0
## ht                    0         0    10     0      0      0     0
## wt1                   0         0     0    10      0      0     0
## number                0         0     0     0     10      0     0
## parity                0         0     0     0      0     10     0
## age                   0         0     0     0      0      0    10
```

Comparison Table

```r
means1 = fit1$coefficients
means2 = fit2$coefficients
means3 = fit3$coefficients
means4 = fit4$coefficients

comparison = data.frame(
  Term          = names(means1),
  Prior_Mean    = round(b0_informed, 3),
  Post_Diffuse  = round(as.numeric(means1), 3),
  Post_Moderate = round(as.numeric(means2), 3),
  Post_Informed = round(as.numeric(means3), 3),
  Post_Tight    = round(as.numeric(means4), 3)
)

# Add a column showing direction match with prior
comparison$Dir_Match = ifelse(
  sign(comparison$Prior_Mean) == 0 |
  sign(comparison$Post_Diffuse) == sign(comparison$Prior_Mean),
  "Yes", "No"
)

print(comparison)
```

```
##          Term Prior_Mean Post_Diffuse Post_Moderate Post_Informed Post_Tight
## 1 (Intercept)       0.00       22.120        22.550        22.120     22.267
## 2   gestation       0.70        0.011         0.011         0.011      0.011
## 3          ht       0.84        1.375         1.367         1.375      1.376
## 4         wt1       0.12        0.059         0.060         0.059      0.059
## 5      number      -1.80       -1.540        -1.530        -1.540     -1.541
## 6      parity       5.30        0.060         0.059         0.060      0.111
## 7         age       0.00        0.037         0.037         0.037      0.029
##   Dir_Match
## 1       Yes
## 2       Yes
## 3       Yes
## 4       Yes
## 5       Yes
## 6       Yes
## 7       Yes
```

Visualization

```r
terms = names(means1)[-1]  # exclude intercept

prior_labels = c("Diffuse\n(zero mean)",
                 "Moderate\n(zero mean)",
                 "Informed\n(diffuse var)",
                 "Informed\n(tight var)")
```

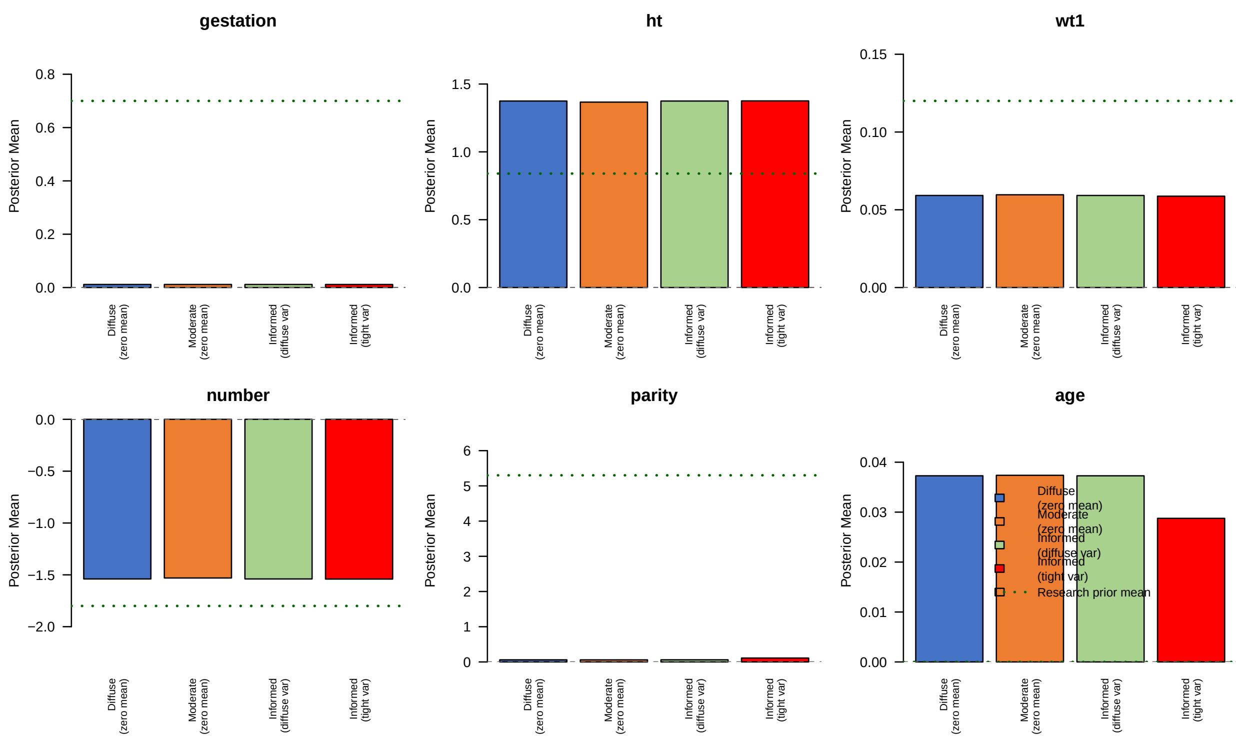
```r
cols = c("#4472C4", "#ED7D31", "#A9D18E", "#FF0000")

par(mfrow = c(2, 3), mar = c(5, 4, 3, 1))

for (term in terms) {
  idx = which(names(means1) == term)
  vals = c(means1[idx], means2[idx], means3[idx], means4[idx])
  prior_val = b0_informed[idx]

  bp = barplot(
    height    = vals,
    names.arg = prior_labels,
    main      = term,
    ylab      = "Posterior Mean",
    col       = cols,
    cex.names = 0.7,
    las       = 2,
    ylim      = range(c(vals, prior_val, 0)) * 1.3
  )
  abline(h = 0, lty = 2, col = "gray40")
  # Mark the research-based prior mean as a reference line
  abline(h = prior_val, lty = 3, col = "darkgreen", lwd = 2)
}


legend("center",
       legend = c(prior_labels, "Research prior mean"),
       fill   = c(cols, NA),
       lty    = c(NA, NA, NA, NA, 3),
       lwd    = c(NA, NA, NA, NA, 2),
       col    = c(rep("black", 4), "darkgreen"),
       bty    = "n",
       cex    = 0.9)
```



Credible Region Test

We test H0: all slope coefficients = 0 vs H1: at least one != 0. With 6 slope coefficients, the chi-square critical value at alpha=0.025 is 14.449.

```
run_cr_test = function(fit, label) {
  b1 = fit$coefficients[-1]
  V1 = fit$post.var[-1, -1]
  cr = as.numeric(t(b1) %*% solve(V1) %*% b1)
  cat(label, "-> test statistic =", round(cr, 4))
  if (cr > 14.449) {
    cat("  (reject H0)\n")
  } else {
    cat("  (fail to reject H0)\n")
  }
}

run_cr_test(fit1, "Prior 1: Diffuse zero-mean     ")
```

```
## Prior 1: Diffuse zero-mean      -> test statistic = 107.0425  (reject H0)
```

```
run_cr_test(fit2, "Prior 2: Moderate zero-mean    ")
```

```
## Prior 2: Moderate zero-mean     -> test statistic = 106.6186  (reject H0)
```

```r
run_cr_test(fit3, "Prior 3: Informed means, diffuse")
```

```
## Prior 3: Informed means, diffuse -> test statistic = 107.0434  (reject H0)
```

```r
run_cr_test(fit4, "Prior 4: Informed means, tight ")
```

```
## Prior 4: Informed means, tight  -> test statistic = 107.4947  (reject H0)
```